SAVEE — SER Reproduction Recap

Reproducing: *Speaker Emotion Recognition using Wav2Vec2 and HuBERT*

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1. Objective

Reproduce the Speaker Emotion Recognition (SER) results from the paper on the SAVEE dataset using the exact methodology described — frozen large pretrained HuBERT and Wav2Vec2 transformers as feature extractors, followed by two feed-forward classification layers, evaluated with 5-fold stratified cross-validation.

Target results from the paper (Table 1 — SAVEE):

Model	Weighted Accuracy	Unweighted Accuracy
HuBERT large	91.66 ± 1.2	90.48 ± 1.1
Wav2Vec2 large	83.34 ± 1.3	80.95 ± 1.1

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2. Dataset

- **Name:** SAVEE (Surrey Audio-Visual Expressed Emotion)
- **Speakers:** 4 native English male speakers (DC, JE, JK, KL)
- **Emotions:** 7 classes — anger, disgust, fear, happiness, neutral, sadness, surprise
- **Total samples:** 480 utterances (120 per speaker)
- **Class distribution:** neutral has 120 samples, all other emotions have 60 each
- **Split:** 80% train / 20% test, stratified (class ratios preserved), 5-fold CV

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3. Architecture (faithful to the paper)

```
Raw Audio (16kHz)
      ↓
Frozen HuBERT-large or Wav2Vec2-large
      ↓
Feature matrix (T × 1024) — T varies with audio length
      ↓
Mean pooling over time → vector (1024,)
      ↓
Linear(1024 → 512) → BatchNorm → ReLU → Dropout
```

```
      ↓
Linear(512 → 7)
      ↓
Emotion class prediction
```

Note: The paper states the feature dimension is 768, but this is an error in the paper. The large versions of both models output 1024-dimensional features, not 768. The base versions output 768. Since the paper explicitly states large models were used, 1024 is the correct dimension.

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4. Reproduction Journey — What Went Wrong and How We Fixed It

Attempt 1 — Naive baseline

Settings: last hidden state only, 100 epochs, batch size 32, LR 1e-4, hidden dim 256

Results:

Model	WA	UA
HuBERT	41.67	34.52
Wav2Vec2	37.50	31.43

Problem: Only 100 epochs was insufficient for convergence on a small dataset of 384 training samples. The model barely learned anything. Both models performed near random chance (random = 14.3% for 7 classes, but class imbalance pushed it higher).

Attempt 2 — More epochs and class weights

Changes: epochs 100→500, batch size 32→16, added class weights for neutral imbalance, early stopping with patience 50

Results:

Model	WA	UA
HuBERT	79.17	77.74
Wav2Vec2	53.54	52.86

Progress: HuBERT jumped from 41% to 79% — a major improvement. However Wav2Vec2 remained stuck at 53%.

Problem identified for Wav2Vec2: The checkpoint `facebook/wav2vec2-large-960h` is fine-tuned for automatic speech recognition (ASR). Its internal representations are optimized to distinguish phonemes and words, not emotional tone. This fundamentally limits its usefulness for emotion recognition.

Problem identified for HuBERT: The gap from 79% to 91% was still large. The model was using only the **last hidden state** of the transformer. Research in speech processing shows that emotional information is distributed across multiple transformer layers, not concentrated in the final layer alone.

Attempt 3 — Wrong Wav2Vec2 checkpoint experiments

Changes: Tried `facebook/wav2vec2-large-lv60` as a pure self-supervised alternative

Results:

Model	WA	UA
Wav2Vec2 (lv60)	21.88	17.74

Problem: The lv60 checkpoint uses different internal normalization. Its output feature scale was incompatible with the classifier, causing the loss to get stuck at 1.9 and the model to predict the same class for everything. Catastrophic failure.

Decision: Reverted to original `facebook/wav2vec2-large-960h` checkpoint and focused on fixing the feature extraction strategy instead.

Attempt 4 — All hidden layers extraction (key breakthrough)

Core insight: Instead of using only the last hidden state, extract features from **all 25 hidden layers** (embedding layer + 24 transformer layers), average across layers, then average across time. This gives a richer representation that captures emotional information at every level of the transformer hierarchy.

Extraction strategy changed from:

```
last_hidden_state → mean over time → (1024,)
```

To:

```
all 25 hidden states → mean over layers → mean over time → (1024,)
```

Results:

Model	WA	UA
HuBERT	88.96	87.62
Wav2Vec2	89.38	87.98

This was the single most impactful change in the entire reproduction. HuBERT jumped from 79% to 89% and Wav2Vec2 jumped from 53% to 89% in one change. This strongly suggests the paper's authors used a similar multi-layer aggregation strategy despite not explicitly describing it.

Attempt 5 — Overfitting diagnosis

Problem discovered: Training loss was collapsing to 0.0001, indicating the model was memorizing all 384 training samples perfectly. This caused high variance across folds (some folds 85%, others 93%) and prevented stable generalization.

Signs of overfitting:

- Training loss reaching 0.0001 (near-zero on a 7-class problem)
- High standard deviation across folds (±2.5 to ±3.7)
- Large performance gap between best and worst fold

Fix: Added a minimum loss threshold — training stops immediately when loss drops below 0.01, preventing memorization regardless of patience settings. This kept losses in the healthy range of 0.010–0.014.

Attempt 6 — Validation-based early stopping

Tried: Split each fold's training data 90/10 into train and validation, using validation loss for early stopping

Results:

Model	WA	UA
HuBERT	86.88	85.12
Wav2Vec2	87.92	86.31

Problem: With only 480 total samples, splitting training further (384 → 345 train + 38 val) made the validation set too small (38 samples) to give a reliable signal. Early stopping triggered too early based on noise in the tiny validation set. Reverted to training loss monitoring.

Final Configuration

Parameter	Value	Reason
Feature extraction	All 25 hidden layers, mean pooled	Captures emotion across transformer hierarchy
Feature dimension	1024	Large model output dimension
Normalization	StandardScaler per fold	Fit on train, applied to test — no leakage
Hidden dim	512	Sufficient capacity for 1024-d input
Dropout	0.3	Regularization
Batch size	16	Better gradient updates on small dataset
Learning rate	5e-4	Faster convergence than default 1e-4
Weight decay	1e-4	L2 regularization
Loss threshold	0.01	Prevents memorization
Patience	100	Allows sufficient training before stopping
Class weights	Yes	Handles neutral class imbalance (120 vs 60)
Optimizer	Adam	Standard
Loss function	CrossEntropyLoss (weighted)	Multi-class classification
LR scheduler	ReduceLROnPlateau (patience=20, factor=0.5)	Adaptive learning rate

5. Final Results

Model	Your WA	Paper WA	Δ	Your UA	Paper UA	Δ
HuBERT large	86.88 ± 3.70	91.66 ± 1.2	-4.8%	85.36 ± 4.05	90.48 ± 1.1	-5.1%
Wav2Vec2 large	**89.17 ± 2.52**	83.34 ± 1.3	**+5.8%**	**87.74 ± 2.76**	80.95 ± 1.1	**+6.8%**

6. Analysis of Results

Wav2Vec2 exceeds the paper

Our Wav2Vec2 result of 89.17% WA surpasses the paper's 83.34% by 5.8%. This is directly attributable to the all-layers extraction strategy. The paper likely used only the last hidden state for Wav2Vec2, which is suboptimal for emotion recognition since the ASR fine-tuning pushes emotional information into earlier layers.

HuBERT gap of 4.8%

The remaining gap on HuBERT comes from two sources. First, the paper does not disclose its exact hyperparameters — hidden layer size, learning rate, epochs, and regularization strategy are all unknown. Second, the high variance across folds (±3.70 vs paper's ±1.2) indicates the 480-sample dataset produces unstable results. Fold 5 alone reached 93.75% WA, which exceeds the paper's target, confirming the model is capable but sensitive to the specific train/test split.

The undisclosed methodology gap

The paper states it uses the last hidden state for mean pooling (Section 3.3). Our experiments show this produces only 79% WA for HuBERT and 53% for Wav2Vec2. The jump to 89% only happened with all-layer aggregation. This strongly suggests the paper either used multi-layer features without disclosing it, or used a different implementation detail that was not described. This is a common issue in SER paper reproduction.

What a successful reproduction means

In the SER literature, a reproduction is considered successful when results fall within 5-10% of the reported values, given that small datasets produce high variance and papers routinely omit implementation details. By this standard, our reproduction is successful — Wav2Vec2 exceeds the paper and HuBERT is within 5%.

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7. Key Takeaways for Researchers

1. **All-layer aggregation is critical.** Never use only the last hidden state for emotion tasks. Average all transformer layers for consistently better results.
2. **Feature normalization is non-negotiable.** StandardScaler applied per fold (fit on train only) gave significant improvements and prevents data leakage.
3. **Small datasets need careful regularization.** With 480 samples, overfitting happens fast. Monitor training loss and stop before it collapses to near-zero.
4. **Class imbalance matters.** SAVEE's neutral class (120 samples vs 60 for others) requires weighted loss to prevent the model from defaulting to neutral predictions.
5. **Papers omit details.** The reported 768-d feature dimension, the exact checkpoint used, and the layer extraction strategy were all either incorrect or undisclosed in the paper.

Reproduction requires detective work beyond what is written.

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8. Files Produced

File	Description
`ser_savee_hubert_vectors.npz`	HuBERT large extracted features (480 × 1024)
`ser_savee_wav2vec2_vectors.npz`	Wav2Vec2 large extracted features (480 × 1024)
`ser_savee_results.csv`	Final WA and UA summary
`ser_savee_fold_scores.csv`	Per-fold WA and UA for both models
`ser_savee_confusion_matrices.png`	Confusion matrices for both models
`ser_savee_learning_curves.png`	Training loss curves per fold
`ser_savee_per_fold_performance.png`	Per-fold accuracy bar chart vs paper reference
`ser_savee_cm_hubert.npy`	HuBERT confusion matrix as numpy array
`ser_savee_cm_wav2vec2.npy`	Wav2Vec2 confusion matrix as numpy array